

SHETH LUJ COLLEGE

Aim: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

CLI

Output

```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/Cyber Security/cs_3.py =====
Enter the message: lets goo parliament
Enter the secret key: 2

Generated MAC: cdfb2337d5a275c75099eb88b33fde00d6119bb4c09ee7aa311ad183013475b9

--- Verification ---
Enter received message: lets goo parliament
Enter secret key: 2
Generated MAC at Receiver: cdfb2337d5a275c75099eb88b33fde00d6119bb4c09ee7aa311ad183013475b9
MAC Verified: Message is Authentic and Unchanged.

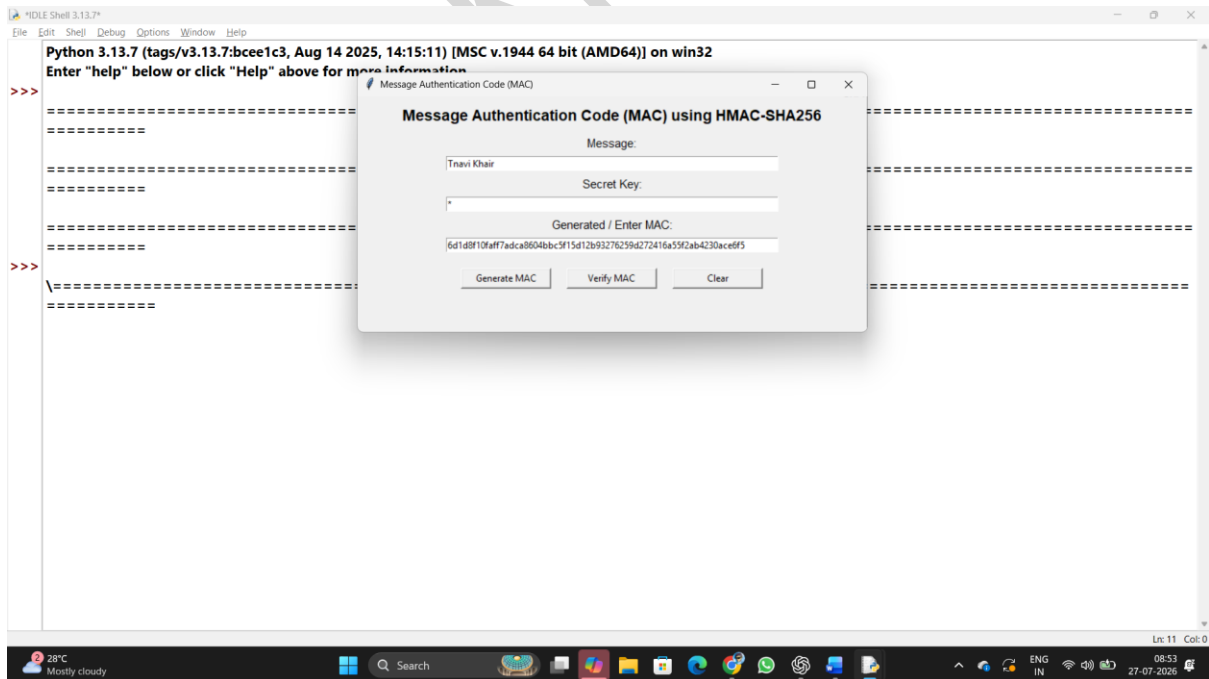
>>>
===== RESTART: C:/Daniyal khan/Cyber Security/cs_3.py =====
Enter the message: I am in TYCS
Enter the secret key: 1

Generated MAC: 73efd88de99784dcc422088bed8977cb27f75be5b2a16f3f41ece216e2f40bea

--- Verification ---
Enter received message: I was in FYCS
Enter secret key: 2
Generated MAC at Receiver: b233e1a077bb93c0c6efdc2c6fdcd0193f1994c2affad862baa530285bcdcae5
MAC Verification Failed: Message has been Modified or Key is Incorrect.

>>> |
```

GUI



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